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Effects of third-party information on the demand for more sustainable consumption: A choice experiment on the transition of winter tourism

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ABSTRACT

Sustainability transitions involve a dynamic interaction of regulation, innovation, and preferences for new products. In this study we examine consumer preferences as a potential driver of supply side innovations in the winter tourism industry. We use a novel survey approach that allows for heuristic decision-making based on information from third parties. Specifically, we investigate consumer preferences when (i) *only* factual information and (ii) factual information *and* environmental NGO or consumer recommendations are available. The third-party information significantly affected the destination choices in about one third of the examined choice sets. The results indicate a willingness to pay for some of the examined sustainability attributes, and they point to specific differences among different tourist groups. Regarding methodology we suggest that survey experiments involving third-party information are useful for the study of choices about product or policy innovations in dynamic markets where standard assumptions on preferences and rationality may not apply.

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1. Introduction

The transition to a more sustainable economy involves complex interactions of environmental scarcities and their perception, public regulation, private innovations, and consumer preferences, among other factors (Witt, 2011). A better understanding of these interactions is important for identifying specific barriers to transitions and for proposing solutions to overcome those barriers (Jacobsson and Bergek, 2011). One potential barrier to sustainable innovations and market formation is the lack of knowledge about consumer demand. If producers are uncertain about consumer demand they are naturally reluctant to innovate and invest in more sustainable products and services.

In the marketing and environmental economics literature, one way to approach uncertain consumer demand is to collect evidence on consumer preferences using stated preference surveys (Louviere et al., 2000). An important advantage of survey based approaches in the context of product innovations is that they allow researchers to explore preferences for products that do not yet exist. A disadvantage is, however, that the traditional approaches in survey research rely on assumptions far beyond the usual one that preferences are truthfully reported: preferences for (novel) products are well defined and respondents systematically process the relevant technical product information provided by the researchers (McFadden, 1999). Limitations of these assumptions which have become widely known under the term of 'bounded rationality' (Simon, 1957) are particularly relevant to the study of innovations and societal transitions where preferences may not always be well defined and technological characteristics of products may be only one among many factors determining consumer decisions (e.g. Witt, 2010).

In markets for novel products, including more sustainable product varieties, boundedly rational consumers may not only rely on individual preferences and systematic processing of technical information. They may also heavily rely on 'heuristic' decision-making (Tversky and Kahneman, 1974), which involves the use of simple decision rules in order to reduce mental effort in dealing with complex or unfamiliar situations. For instance, consumer decision-making heuristics may be based on recommendations from friends or on the imitation of potentially better informed individuals (Slembeck and Tyran, 2004). Reliance on heuristic decisions can be sensible for several reasons. Consumers may lack consistent preferences for unfamiliar new products, product information may be limited, or consumers may not have the time or cognitive capacity to process the information. Furthermore, products like fashion items may serve as signals of group membership while technical characteristics of the products are less important (Witt, 2010). Under such conditions, network interactions among agents become an important point to consider (Van den Bergh et al., 2011). Social interactions may affect the preferences and choices of consumers, and consumer networks may in turn influence potential innovators in the supply chain (Jacobsson and Bergek, 2011). Hence, in the analysis of consumer demand using surveys, it may not be appropriate to use standard economic models and traditional survey approaches that do not account for heuristic decision-making and social interactions among consumers.

The aim of this study is to explore consumer demand for product innovations when standard economic assumptions on preferences and decision-making may not apply and consumers may choose to rely at least partly on heuristic decision-making. In decisions about sustainable products, a key type of heuristic information is advice from (credible) third parties such as environmental organizations or more experienced consumers who do not have a financial stake in the products. We thus design and implement a choice experiment to investigate consumer choices under alternative sets of such information. The control treatment in the experiment follows a standard choice experiment design. In two separate treatments, we additionally provide (researcher-defined) third-party information in the form of "recommendations from an environmental NGO" and "Consumer recommendations".

The application of the experiment is to winter tourism services. Specifically, we investigate decisions about winter tourism destinations with alternative services and sustainability attributes. The attributes include both "green" adaptations including the substitution of non-renewable by renewable energy and behavioral adaptations such as offers for snow-independent activities. Our objective is to explore whether consumer demand for more sustainable winter tourism services might warrant private investments in more sustainable tourism offers. A second, more fundamental objective is to investigate how heuristic sustainability information may affect choices about sustainable consumption and how this may be relevant to the analysis of demand using surveys.

The following two sections provide background on the market for sustainable ski tourism and on the role of third-party information in preference elicitation. Section 4 presents the methods. Section 5 reports the results, followed by a discussion and conclusions.

2. Background on the market for more sustainable winter tourism

Tourism is the main industry in many parts of the European Alps, with up to 90% of local economic dependency in certain Alpine regions and about equal shares of summer and winter tourism for the Alpine average (CIPRA, 2006). Climate change has been discussed as the major threat for winter tourism (Abegg, 1996; Bürki, 2000; Elsasser and Bürki, 2002; Zemp et al., 2006; IPCC, 2007; OECD, 2007; Endler et al., 2009) and tourism in general (Scott, 2011). Unusually warm recent winters such as the season of 2006/07 give an impression of what may be average conditions in the future (Beniston, 2007; Uhlmann et al., 2009; Falk, 2010). The current emphasis on maintaining services through the production of snow (Yang and Wang, 2011) is seen as neither ecologically sustainable (CIPRA, 2006; Müller and Weber, 2008) nor economically viable in the long term (OECD, 2007; Töglhofer et al., 2011). There is a need for alternative strategies and a transition to sustainable business models including a diversification of tourism services and the mitigation of climate change impacts (Loibl and Walz, 2010; McKercher et al., 2010).

Within the ski tourism industry, mitigation of greenhouse gas emissions and consumer adaptation have been neglected in the discussion about possible responses to climate change (OECD, 2007; Wolfsegger et al., 2008). However, the sustainable (ski) tourism market could become more important in the future (Abegg et al., 2008; Roth et al., 2008; Miller et al., 2010). Behavioral adaptation including snow-independent winter activities and mitigation including investments in energy efficiency and renewable energy could reduce negative impacts from tourism on the environment and decrease the dependency on the most vulnerable Alpine tourism product, downhill skiing (CIPRA, 2006). Furthermore, new opportunities for private investors can arise if consumers valued such services and changed their behavior accordingly (Miller et al., 2010). Evidence of a substantial market demand for more sustainable consumption in winter tourism could help overcome the existing corporate reluctance to establish more sustainable business models based on behavioral adaptation and mitigation (such as in tourism in general, see Miller et al., 2010). Further preconditions for change include alternative models of destination governance (Beritelli et al., 2007) and guidance and support from policy makers for the corporate adaptation process (Hoffmann et al., 2008).

To date, there is a lack of data on customer demand for more sustainable (Alpine) tourism (Hamilton et al., 2005; Falk, 2010; Konu et al., 2011). Existing survey research shows, however, that environmental sustainability in Alpine tourism was considered “important” or “very important” by 57% of Swiss tourists, while 74% were “in general” willing to pay more for a product or service that has a lower negative impact on the environment – but only 56% were willing to accept “less comfort” in such a product (Müller et al., 2001). Technical snow making, despite of its resource intensiveness, was considered “acceptable” by 50–60% of the guests in three Swiss winter destinations (Davos, Scuol and Braunwald), mainly due to its economic importance for the region (Teich et al., 2007) and by even 90% after the very warm and dry winter 2006/07 (Luthe, 2009). Climate change is considerably lowering the demand for Alpine tourism in winter (Abegg, 1996; König, 1998; Bürki, 2000; Scott et al., 2006; Siomkos et al., 2006; ÖIR, 2007; Unbehaun et al., 2007; Luthe, 2009; Pickering et al., 2009; Töglhofer et al., 2011) but potentially increasing demand in summer (Serquet and Rebetez, 2011).

This paper focuses on the demand side, acknowledging, however, that the study of supply side considerations is equally important (see e.g. Hoffmann et al., 2008; Loibl and Walz, 2010). Our goal is to contribute to the understanding of consumer preferences for more sustainable (tourism) services and to explore, based on the results of our case study, whether less technical or resource intensive adaptation to climate change in alpine winter tourism can indeed become a promising business case.

3. Background on preference elicitation and the role of third-party information

As noted in the introduction, traditional preference elicitation research has maintained the assumption that survey respondents know their preferences and systematically process relevant technical

information even if choice tasks involve complex and unfamiliar alternatives (Fischhoff, 1991; McFadden, 1999). Accordingly, researchers have relied on survey protocols that require respondents to make their choices in isolation and based on purely technical information about alternative products or policies (Mitchell and Carson, 1989; Louviere et al., 2000).

These approaches are increasingly questioned by researchers who carefully examine how individual decisions are actually made in various decision environments (Tversky and Kahneman, 1974, 1981; Gregory et al., 1993; Schkade and Payne, 1994; Ariely et al., 2003; Fischhoff, 2010). A large body of experimental decision research now shows that choices about unfamiliar alternatives are not always made in ways consistent with traditional economic theory and preference elicitation. Instead, decision-makers often rely on simple heuristics using contextual information conveyed by market prices (Ariely et al., 2003), observation and imitation of other, possibly better informed individuals (Slembeck and Tyran, 2004), issue positions of political parties (Druckman, 2001) or signals conveyed by default rules (Sunstein and Thaler, 2003; Thaler and Sunstein, 2008). Absent such contextual information, choices are often found to be inconsistent and highly sensitive to arbitrary framing (Tversky and Kahneman, 1981; Ariely et al., 2003). These issues are increasingly recognized also in the economic valuation literature (Munro, 2009).

Dealing with the implications of heuristic decision-making, however, the traditional stated preference research has primarily focused on the identification of resulting biases. Only recently have researchers begun to study how decision environments can be designed to support the kind of heuristics that people use successfully in many actual consumer or political decisions (Payne et al., 1999; Sunstein and Thaler, 2003). One way to address the issue of heuristic decision-making is based on the idea of 'context matching' (Payne et al., 1999). Researchers have started to consciously provide contextual information in laboratory and survey experiments in order to 'match' the context of decisions in more natural or more relevant environments in which rich contextual information is often available. One approach, initiated by Macmillan et al. (2002), allows for group discussion among respondents of environmental valuation surveys. Several studies suggest that group discussion enhanced the consistency of stated preferences for environmental policies (e.g. Macmillan et al., 2002; Alvarez-Farizo et al., 2007, 2009). In another approach, Huffman et al. (2007) studied choices about genetically modified food under alternative sets of information (pro-environment, pro-industry and neutral perspectives). The authors found that relatively uninformed respondents were sensitive to the information. A similar approach was used to study preferences for proposed environmental policies in the presence *vs.* absence of information about the positions of political parties and interest groups using choice experiments (Schläpfer et al., 2008) and contingent valuation (Schläpfer, 2011). In line with observational research in political science, access to party positions increased choice consistency even among well-educated individuals. Furthermore, heuristic decision-making based on advice from other individuals or groups has been examined in medical decision research (Haward et al., 2008).

The present study builds on that previous research and examines the role of third-party information on consumer choices in the context of a transition towards more sustainable products and services. The explicit consideration of third-party information takes into account that group influences, learning and social coordination may play an important role in such transitions (Witt, 2010, 2011; Jacobsson and Bergek, 2011). Experiments with third-party information thus provide an opportunity to explore consumer preferences for alternative products in social decision environments where specific information and information sources are available to the decision-makers.

In choices about more sustainable winter tourism services, which are the focus of this study, recommendations from environmental NGOs or from more experienced consumers are one potentially relevant type of third-party information. We therefore examine the demand for sustainable winter tourism services in a choice experiment in which access to such recommendations is experimentally controlled.

4. Methods

4.1. Choice experiment design

The basic choice experiment design follows standard procedures (Louviere et al., 2000). In attribute-based stated preference questions, a good or product, in our case a ski destination, is decomposed into

Table 1
Attributes and levels.

Attribute name	Definition	Levels
HOURS	Time and mode of travel to ski area	Less than 2 h 2–4 h 4–8 h More than 8 h (including plane)
SMALL	Size and character of area	Small and authentic: 20 km groomed runs Large and world class: 100 km groomed runs
RENEW	Percentage of renewable energy used to power lifts	0% 100%
ALT	Availability of snow independent alternatives	ALT1: downhill without snow[a] ALT2: mountain coaster in ski area ALT3: snow dome nearby ALT4: soft alternatives, e.g. hiking
FLEX	Transferability of ski passes	Non transferable Fully transferable throughout season
EXP	Expansion of ski area as part of technical adaptation to extend terrain horizontally/vertically	EXP1: no expansion EXP2: e. to higher elevations EXP3: e. onto glacier EXP4: e. by connection of two areas
GROOM	Percentage of runs which are groomed regularly	50% of total runs are groomed 100% of total runs are groomed
SNOW	Snow reliability achieved by technical snow making in days	100% – 6 days out of 6 80% – 5 days out of 6 65% – 4 days out of 6 50% – 3 days out of 6
PRICE	Price of a day ski pass in €	34, 42, 50, 58

[a] Downhill without snow includes downhill biking and sledging on rails.

k singularly and precisely specified attributes. The combination of the attributes, each with two or more alternative levels, constitutes the alternative profiles (ski destinations). Two or more alternative profiles constitute a choice set. In the present study, the choice sets consisted of two ski destinations, together with an opt-out option for cases where a consumer would not take the ski trip at all rather than choosing one of the offered alternatives. The respondent thus had three options available in a choice set – "ski area A", "ski area B", and "none of them".

Profiles were characterized by 8 ski area attributes and a price attribute, which were the costs of a day ski pass. The choice of relevant attributes and attribute levels (Table 1) was based on interviews with industry experts (Luthe, 2009) and reflects the expected importance of these factors in future ski area choices. Travel time (attribute name HOURS) is a cost factor but also an environmental sustainability attribute since longer travel is associated with higher resource use. Snow guarantee (SNOW), which is usually based partly on artificial snow making, and size of ski area (SMALL) are well known determinants of customer satisfaction. Adaptation strategies include various ways of expanding the ski area (EXP1–4) and offering a variety of alternatives to skiing (ALT1–4). Relevant mitigation strategies include the use of electricity produced by renewable energy (RENEW) and reduced grooming of the slopes (GROOM). Flexible ski passes (FLEX) allow customers to spontaneously decide about alternative (non-skiing) activities. The price of the one-day ski pass (PRICE) was described by four levels ranging from typical prices in Alpine ski areas (34€ and 42€) to somewhat higher levels (50€ and 58€) in order to explore also higher willingness to pay values.

The fractional factorial design of the choice experiment was generated using the program Gosset (Hardin and Sloane, 2003). We specified an *i*-optimal design, which minimizes prediction variance, and a target number of 64 ski area profiles. Profiles can be combined to sets randomly or systematically (Louviere et al., 2000). Street et al. (2005) propose various strategies for combining options to choice

Table 2
Evaluation of attribute levels in the definition of environmental NGO and consumer recommendations.

Variable	Negative evaluation	Positive evaluation
HOURS	>8	<2
SMALL	–	Small
RENEW	0	100%
Alternative offers	ALT2, ALT3	ALT4
EXP	EXP3	EXP1
FLEX	–	Transferable
GROOM	–	50%
SNOW	100%	50%

Note: see Table 1 for description of variables and levels.

sets. From groups of eight profiles we systematically paired the first with the second profile, the second with the third and so forth. The eighth profile was then paired with the first profile (see Section 5; Table 4). The resulting sixty-four choice sets were checked for combinations with dominant options (Louviere et al., 2000). We identified one dominant choice set, number 45, which was changed by switching profiles 45 and 44 (see Table 4). In order to keep the choice task manageable, the resulting sixty-four choice sets were allocated to eight blocks including choice sets 1 through 8, 9 through 16, etc. The 64 profiles are presented in the Appendix A (Table A.1).

4.2. Information treatments

Corresponding with the aims of the study the experimental design was extended by allocating the respondents to different information treatments. Three different information treatments were applied:

1. Factual information only ("Control")
2. Factual information and recommendations "from a well-known environmental NGO" ("NGO information")
3. Factual information and recommendations "from other customers who already visited the ski areas" ("Consumer information").

The "NGO information" treatment was chosen to explore the effects of an environmental "certification" or other form of endorsement by a trusted third party that is not affiliated with the ski area. This kind of information could be part of a marketing strategy of more sustainable ski areas if it showed to have a significant effect on consumer choice. The "consumer information" is applied to test whether any observed effects can be attributed to the source or whether a more generic source of consumer recommendations may produce similar effects.

The recommendations were assigned based on a qualitative assessment of the sustainability of the ski area options. The recommendations could take any of the three following forms:

(i) positive recommendation of one of the ski areas (if one is more sustainable than the unless both are relatively unsustainable);
(ii) negative recommendation of both ski areas (if both are relatively unsustainable); or
(iii) no particular recommendation (if no positive or negative attribute levels occur or if positive and negative levels are similar in frequency).

The qualitative assessment of sustainability is based on an evaluation of the individual attribute levels as positive, negative or neutral. Positive, negative and neutral evaluations were based on an environmental management scheme "Auditing in Ski Areas" (Pröbstl et al., 2003), recommendations and publications of the Board of Environment and Sustainable Snow Sports Development of the German Ski Federation (Lauterwasser, 1989) and additional literature on environmental aspects of ski tourism (see Luthe, 2007).

Table 3
Descriptive statistics.

Variable	Categories	Mean	Std. dev.	Test statistic[a]	*P*-value	*N*
Gender	0 = Male, 1 = female	0.33	0.47	$\chi^2 = 0.47$	0.79	2389
Age	Age in years	37.0	12.4	$F = 0.49$	0.61	2389
Income	1 = <1100€, 2 = 1100–2000€, 3 = 2100–3000€, 4 = 3100–4000€, 5 = 4100–5000€, 6 = >5000€	3.30	1.43	$\chi^2 = 0.69$	0.71	2269
Travel time to nearest ski area	1 = <1 h, 2 = 1–2 h, 3 = 2–4 h, 4 = 4–8 h, 5 = >8 h plane, 6 = >8 h no plane	2.37	1.32	$\chi^2 = 5.14$	0.08	2389
Skills	1 = Professional, 2 = very good, 3 = good, 4 = fair, 5 = beginner	2.17	1.22	$\chi^2 = 0.31$	0.86	2389
Days of skiing (annually)	1 = <1 week, 2 = 1–2 weeks, 3 = 2–4 weeks, 4 = >4 weeks	2.58	0.88	$\chi^2 = 0.23$	0.89	2389
Knowledge (climate change)	1 = Knowledge, 2 = no knowledge, 3 = do not know	1.14	0.41	$\chi^2 = 1.67$	0.44	2389

[a] Tests for equality of treatment subsamples: gender: chi-square test, age: *F*-test, other variables: Kruskal–Wallis *H*-tests ($df = 2$).

Negative evaluations of attribute levels were based on the following considerations. Travelling to and from ski destinations accounts for the majority of greenhouse gases of an average 1-week ski holiday in an Alpine ski destination (SLF, 2006; Mountain Riders, 2007). The level "more than eight hours" was assigned a negative evaluation. Expansion on glaciers was considered unsustainable as it threatens fragile alpine environments. Full (100%) snow guarantee was considered unsustainable because it typically relies on excessive technical snow making (CIPRA, 2006).

Positive evaluations were assigned as follows. A travel distance of less than 2 h implies that traffic externalities are limited. Soft mountain sport alternatives such as hiking or biking use fewer resources than technical ones. Small and authentic ski areas are more supportive of local traditions and were therefore assumed to impact less negatively on landscapes. Flexibility of ski passes enable tourists to switch to alternative activities when snow conditions are inadequate (and come back for skiing another day). Both grooming and snow making are resource-intensive and produce emissions in fragile environments. Therefore, reduced grooming and snowmaking were evaluated positively. Ski areas without positive or negative recommendations were accompanied by a neutral statement.

The evaluation of the attribute levels used in the definition of the recommendations is summarized in Table 2. In constructing the recommendations, the "strongly" negative attribute levels ">8 h travel time", "mountain coaster in ski area" (ALT2) and "expansion onto glacier" (EXP3) and the "strongly" positive attribute level "100% renewable energy" received twice the weight of the other negative and positive attribute levels.

The wording of the recommendations is presented in the Appendix A (Table A.2). The formulation of the recommendations reflects existing communication strategies in the North American ski resorts Aspen and Vail (Colorado) which have already been marketing sustainable ski area management in recent years (M. Hamilton from Aspen Skiing company, personal communication). The respondents

Table 4
Tests of information effects on destination choice.

No.[a]	Profiles	Descriptive information and NGO recommendations[c]			Descriptive information and consumer recommendations[c]			Descriptive information only (control)[c]			Test of NGO effect	Test of consumer effect	Test of NGO *vs.* c. effect
	(A/B)[b]	*N*	%A	%B	*N*	%A	%B	*N*	%A	%B	($\chi^2_{<2>}$)	($\chi^2_{<2>}$)	($\chi^2_{<2>}$)
1	1/2(+)	99	5.8	81.8	93	14.7	62.4	97	14.4	54.6	16.7***	1.7	8.7*
2	2(+)/3	99	76.8	2.0	93	60.6	14.9	97	53.6	17.4	17.3***	0.8	11.7**
3	3/4(+)	99	9.9	68.3	93	11.8	75.3	97	15.5	57.7	2.7	7.2*	2.6
4	4(+)/5	99	79.4	2.0	93	77.4	6.5	97	69.0	7.0	4.4	1.9	2.5
5	5/6(+)	99	1.0	81.8	93	9.6	75.3	97	5.0	66.0	7.3*	4.1	7.3*
6	6(+)/7	99	73.7	8.9	93	69.9	14.7	97	58.8	20.6	6.4*	2.6	1.6
7	7(−)/8(−)	99	11.9	63.6	93	10.6	54.8	97	19.4	61.9	2.5	7.4*	2.0
8	8(−)/1(−)	99	70.7	3.0	93	63.8	2.1	97	74.2	6.1	2.1	6.5*	0.9
9	9(+)/10	94	34.0	56.9	107	33.6	58.2	97	21.7	62.9	4.3	5.0	0.5
10	10(+)/11	94	89.4	3.2	107	89.7	1.9	97	82.5	4.1	2.0	2.4	0.3
11	11(−)/12(−)	94	11.5	45.7	107	10.3	55.1	97	9.3	62.9	5.8	1.3	2.2
12	12( )/13( )	94	23.4	34.0	107	22.4	41.1	97	25.5	39.2	1.1	0.3	1.1
13	13( )/14( )	94	30.9	48.9	107	24.3	48.6	97	32.0	49.5	0.1	2.7	1.7
14	14( )/15( )	94	60.6	11.7	107	66.7	6.5	97	68.0	12.4	1.7	3.1	1.8
15	15( )/16( )	94	44.7	16.0	107	43.9	15.0	97	55.7	14.4	2.5	3.2	0.1
16	16/9(+)	94	28.7	44.7	107	32.7	34.6	97	34.0	36.1	1.5	0.2	2.1
17	17( )/18( )	104	22.1	38.5	109	25.7	38.5	102	28.4	47.1	5.3	3.3	0.1
18	18( )/19( )	104	25.5	58.5	109	16.5	63.3	102	20.6	61.8	0.6	0.7	3.6
19	19(−)/20(−)	104	75.0	6.7	109	71.6	4.5	102	76.5	6.9	0.1	2.0	2.1
20	20(−)/21(−)	104	30.8	16.4	109	20.7	25.7	102	32.0	36.9	14.7***	11.3	4.1
21	21(−)/22(−)	104	27.9	29.8	109	21.1	19.1	102	30.4	35.3	1.4	13.9***	6.2*
22	22(−)/23(−)	104	27.9	43.3	109	25.5	36.7	102	29.1	43.7	0.2	3.0	1.7
23	23(−)/24(−)	104	41.4	20.2	109	47.3	12.7	102	55.9	16.5	4.5	3.5	2.2
24	24(−)/17(−)	104	47.1	10.6	109	31.2	11.0	102	48.0	22.6	6.9*	17.7***	6.0*
25	25/26(+)	100	6.6	80.2	96	10.4	78.1	100	10.7	64.0	6.5*	6.3*	1.3
26	26(+)/27	100	89.0	4.0	96	86.5	10.2	100	78.0	5.9	4.6	10.0**	3.1
27	27(−)/28(−)	100	50.0	8.6	96	46.9	9.1	100	59.0	21.0	12.8**	14.4***	0.2
28	28( )/29( )	100	18.0	47.6	96	24.0	41.2	100	39.0	40.0	11.5**	6.7*	1.7
29	29( )/30( )	100	42.0	22.0	96	49.5	19.6	100	55.0	21.0	4.2	1.0	0.9
30	30(−)/31(−)	100	35.0	23.0	96	41.7	17.4	100	53.5	18.0	6.7*	3.2	1.2
31	31(−)/32(−)	100	16.7	39.0	96	24.7	35.4	100	31.0	41.2	7.7*	3.0	1.9
32	32(−)/25(−)	100	51.0	3.9	96	49.0	6.3	100	59.0	6.0	2.2	2.1	0.6
33	33(−)/34(−)	100	23.0	16.0	99	19.2	26.3	100	31.0	26.4	6.7*	4.1	2.4
34	34(−)/35(−)	100	15.6	27.0	99	26.0	22.2	100	36.0	32.0	15.1***	7.8*	3.5
35	35( )/36( )	100	5.9	73.5	99	10.1	71.4	100	2.9	80.0	1.5	4.3	0.9

Table 4 (*Continued*)

No.[a]	Profiles	Descriptive information and NGO recommendations[c]			Descriptive information and consumer recommendations[c]			Descriptive information only (control)[c]			Test of NGO effect	Test of consumer effect	Test of NGO *vs.* c. effect
	(A/B)[b]	*N*	%A	%B	*N*	%A	%B	*N*	%A	%B	($\chi^2_{<2>}$)	($\chi^2_{<2>}$)	($\chi^2_{<2>}$)
36	36( )/37( )	100	57.0	26.5	99	64.7	17.0	100	60.0	26.0	0.4	2.5	2.6
37	37( )/38( )	100	45.0	21.6	99	40.0	22.0	100	52.5	20.8	1.1	3.1	0.3
38	38( )/39( )	100	24.5	57.0	99	20.2	57.6	100	18.0	66.0	1.9	1.7	0.6
39	39( )/40( )	100	64.0	9.9	99	69.3	4.0	100	71.0	12.0	2.4	5.9	2.8
40	40( )/33( )	100	27.7	20.8	99	29.3	13.1	100	43.0	16.0	4.9	5.6	2.1
41	41( )/42( )	100	63.0	15.0	101	63.4	20.8	103	62.1	13.6	0.2	3.4	0.7
42	42( )/43( )	100	26.9	54.0	101	22.8	52.5	103	17.5	54.4	3.9	1.0	2.1
43	43/44(+)	100	21.0	73.8	101	19.6	74.5	103	14.2	72.8	4.5	3.3	0.0
44	44(+)/45	100	89.2	2.9	101	93.1	2.9	103	86.4	3.7	0.3	2.8	0.7
45	45/46(+)	100	20.0	62.0	101	18.8	59.4	103	14.2	49.5	8.3*	5.0	0.4
46	46(+)/47	100	77.0	1.0	101	72.6	2.9	103	62.1	0.0	6.9*	6.6*	1.0
47	47( )/48( )	100	22.8	29.0	101	10.9	39.6	103	10.6	36.5	5.8	0.2	5.7*
48	48( )/41( )	100	17.8	57.0	101	9.9	57.4	103	7.7	65.1	4.8	1.3	3.1
49	49(−)/50(−)	100	11.0	58.0	100	15.0	42.0	89	10.5	69.7	3.2	15.0***	4.3
50	50(−)/51(−)	100	67.0	6.8	100	59.8	3.9	89	74.2	11.8	5.4	14.9***	2.4
51	51(−)/52(−)	100	26.0	26.0	100	31.1	19.0	89	47.2	26.6	12.0**	11.6**	1.7
52	52(−)/53(−)	100	30.7	18.8	100	30.0	13.0	89	41.6	25.8	5.9	12.0**	1.3
53	53( )/54( )	100	21.0	37.0	100	10.7	39.0	89	20.2	55.1	7.5**	13.1***	4.4
54	54( )/55( )	100	51.0	9.0	100	49.0	5.9	89	53.9	21.4	8.3**	14.1***	1.1
55	55(−)/56(−)	100	28.0	16.0	100	15.7	19.6	89	42.9	17.6	5.7	17.3***	4.5
56	56(−)/49(−)	100	11.0	37.0	100	8.7	29.0	89	15.4	39.6	1.3	5.77	2.5
57	57( )/58( )	96	24.5	53.1	104	20.2	51.9	99	23.0	55.0	0.1	0.9	1.0
58	58/59(+)	96	54.0	36.0	104	44.8	39.4	99	53.5	33.3	0.7	1.4	2.2
59	59(+)/60	96	42.7	43.8	104	41.4	43.8	99	36.4	47.5	1.0	0.5	0.0
60	60(−)/61(−)	96	66.7	4.1	104	55.2	2.9	99	78.0	2.0	3.2	12.1**	2.8
61	61(−)/62(−)	96	9.3	49.5	104	3.8	51.0	99	8.9	55.6	0.6	3.6	2.5
62	62( )/63( )	96	26.8	52.1	104	21.0	56.7	99	23.2	59.6	1.1	0.8	0.9
63	63/64(+)	96	38.1	54.2	104	40.0	51.9	99	39.0	52.0	0.3	0.1	0.1
64	64(+)/57	96	82.3	7.2	104	82.7	6.7	99	80.8	8.1	0.1	0.2	0.0

* Significance level: $p < 0.05$.

** Significance level: $p < 0.01$.

*** Significance level: $p < 0.001$.

[a] Choice set number.

[b] See Appendix A for description of profiles 1 through 64. (+): recommended; (−): disapproved; ( ): no recommendation.

[c] The percentage choosing the option "none" is given by the expression $100 - (\%A + \%B)$.

were randomly allocated to the three information treatments, which were orthogonally crossed with the eight blocks of the choice experiment design. This was achieved by always allocating twenty-four consecutive respondents of the online survey to the twenty-four treatment × block combinations.

4.3. Survey procedures

We used an online survey approach to administer the choice experiment. The use of the Internet allowed the relatively complex survey to be presented in a visually attractive and communicative way. A further advantage of this survey mode is the low cost of administration and data compilation for many respondents as needed for the large experimental design of the study. Software packages for web based choice experiments are commercially available, although with limited options for implementing extended experimental designs. We thus developed our own web survey tool with the required design and functions in collaboration with a professional programmer of the private company *Kapthon AG*, Munich.

The survey consisted of three parts. In the first part, seven general introductory questions were asked. The choice sets were presented next. Part three of the survey (questions 17 trough 35) asked about product and sustainability issues, followed by the questions about socioeconomic characteristics. The results of parts 1 and 3 as well as the complete survey questionnaire are published in Luthe (2009). The answering time for the entire survey was about 10–20 min.

The order of attributes in the choice set was randomly changed with each reload to avoid sequencing effects. A click on the info point button ("*i*"; after the attribute name) provided a list and an explanation of the existing attribute levels.

The levels that were different between the two options were highlighted to assist the respondents in their task. The third-party information (NGO or consumer recommendation) was offered in a non-obtrusive way below the choice set.

The survey instrument was pre-tested with about one hundred respondents to check for technical functionality and for usability. The data was analysed and checked for unexpected effects, for example a dominance of one of the two ski area options (Louviere et al., 2000). One third of the respondents were interviewed about their understanding of the choice tasks and overall experience with the survey. Based on these interviews some additional information on the attributes was included in the text of the info buttons in order to improve understanding. Furthermore, the visual setup of the two ski area options was changed by aligning the attribute names on the left side of the two ski area profiles rather than between them.

4.4. Selection of respondents

The main survey was conducted between January and May of the unusually warm winter of 2006/2007. In order to encourage participation, the survey was combined with a lottery in which participants could win one of 100 day ski passes. Respondents were recruited by collecting a total of 2166 e-mail addresses during short interviews in ski resorts (Arlberg and Zillertal, Austria; 764 respondents/625 valid responses as defined below), where a group of 12 students randomly asked people in lifts and restaurants of the ski area, and in bars, hotels, parking lots, bus stops and the pedestrian areas in the towns for short interviews. Further respondents were recruited by encouraging participation in ski area newsletters with 15,330 subscribers (Oberstdorf and Zugspitze, Germany; Ötztal, Austria; 951/699), in the member magazine of the German Ski Federation with 232,000 subscribers (511/423), and in a radio broadcast (SWR3; 672/449). Additional responses came from students at the University of Freiburg who were recruited through a university newsletter (84/60), from members of the NGO CIPRA (Commission Internationale pour la Protection des Alpes) Switzerland/Liechtenstein who were also recruited through a newsletter (32/20) and from individuals who received the link from other respondents or who did not report how they were recruited (146/113). The respondents were approached in a purposely neutral way. In the ski areas where the interviewers collected e-mail addresses they just asked the respondents to participate in a study about the future of skiing and winter tourism. Likewise in the e-mail newsletter, magazine and broadcasts, potential respondents

Table 5
Multinomial Logit models for the three information treatments.

Attribute	NGO information		Consumer information		Control	
	Coefficient	*t*-value	Coefficient	*t*-value	Coefficient	*t*-value
HOURS	−0.163***	−27.4	−0.149***	−25.6	−0.136***	−23.8
SMALL	−0.489***	−12.7	−0.433***	−11.3	−0.363***	−9.7
RENEW	0.706***	16.1	0.670***	15.6	0.561***	13.4
ALT1	−0.013	−0.3	−0.099**	−2.1	−0.041	−0.9
ALT2	−0.609***	−9.7	−0.723***	−11.5	−0.254***	−4.2
ALT3	−0.259***	−3.0	−0.429***	−4.9	−0.267***	−3.2
FLEX	0.351***	8.1	0.380***	8.9	0.364***	8.6
PRICE	−0.004***	−2.8	−0.008***	−5.2	−0.009***	−6.1
EXP1	0.100**	2.1	0.098**	2.1	−0.010	−0.2
EXP2	−0.253***	−3.6	−0.199***	−2.9	−0.170**	−2.4
EXP3	−0.603***	−8.8	−0.413***	−6.2	−0.259***	−3.9
GROOM	0.074*	1.7	0.042	1.0	0.052	1.2
SNOW	0.257***	19.4	0.258***	19.8	0.313***	24.0
N (choices)	6344	6472	6296			
Log-likelihood	−6074.0		−6281.9		−6070.4	
Log-l. restricted	−6969.6		−7110.2		−6916.8	
McFadden adj. R^2	12.9		11.7		12.2	

* Significance level: $p < 0.1$.
** Significance level: $p < 0.05$.
*** Significance level: $p < 0.01$.

were simply asked to participate in a survey on 'the future of skiing'. The online survey used a neutral design including only one picture in the header, which showed a mountain silhouette.

In total, 3160 participants filled out the online questionnaire. Valid responses were defined as follows: (a) all questions were completed (except for the income and age questions which were not required); (b) the respondent's IP address was unique; and (c) the time stamp indicated that the time to answer the eight choice set questions was at least 60 s. (Forty questionnaires did not pass (c); it is likely that these respondents just clicked through the questionnaire to take part in the lottery.) The 120 respondents who did not report income (see Table 3) were not excluded from the dataset to avoid a selection bias. Filtering according to these criteria resulted in a total of 2389 valid questionnaires (19,112 choice sets responses).

4.5. Statistical analyses

The experimental design allows us to examine a number of hypotheses regarding the effects of the NGO and consumer information on consumer choices involving environmental externalities.

We used chi-square tests to examine the effects of the information treatments (NGO information *vs.* control, consumer information *vs.* control and NGO information *vs.* consumer information) on the distribution of responses in each of the 64 choice sets. We report these extensive disaggregated analyses because they provide simple and clean tests of our hypotheses regarding the effects of information. In contrast, the interpretation of effects in the following multinomial choice models rely on additional assumptions, in particular regarding an orthogonal experimental design, the interpretation of the opt-out option, the functional form (linearity) of the attribute effects, and a zero income effect (Louviere et al., 2000).

Second, we used simple multinomial logit models as implemented in Limdep (command NLOGIT; see Greene, 2003) to further explore how the third-party information affects the pattern of results. The simple modeling approach appears justified given that many issues remain unresolved even in the more advanced analytical approaches (Hoyos, 2010) and that gains from improved statistical analysis may be small compared to gains from using behaviourally more realistic assumptions in generating the data (McFadden, 1999; Ariely et al., 2003; Sunstein and Thaler, 2003). The model specification assumes that the utility of an individual t from choosing alternative j (j = A, B or "no choice") is given by: $U_{jt} = \beta' \mathbf{x}_{jt} + \varepsilon_{jt}$. The nonrandom component is a function of the choice attributes $\mathbf{x}_j$, and the random,

Table 6
Willingness to pay for ski area characteristics.[a]

Attribute (characteristic)	Reference (for nominal variables)	NGO information	Consumer information	Control
HOURS (h of travel time)		−37.3	−19.1	−14.8
SMALL (20 km of runs)	Large (100 km)	−112.1	−55.2	−39.6
RENEW (100% ren. energy)	50% Ren. energy	161.8	85.4	61.2
Alternative offers				
ALT1 (snow-indep. d.hill)	ALT4 (only soft mountain sports)	*−2.9*	−12.7	*−4.5*
ALT2 (mountain coaster)		−139.5	−92.3	−27.7
ALT3 (ski dome)		−59.5	−54.7	−29.2
FLEX (flexible ski pass)	Not flexible	80.4	48.5	39.8
Recent or planned expansion of area				
EXP1 (no expansion)	EXP4 (expansion by combination of two small areas)	22.9	12.4	*−1.1*
EXP2 (to higher elevation)		−58.0	−25.3	−18.6
EXP3 (onto glacier)		−138.2	−52.7	−28.3
GROOM (100% groomed)	50% groomed	17.1	*5.3*	*5.6*
SNOW (snow guarantee d./w.)		59.0	32.9	34.1

Note: interpretation (example): under the NGO information treatment an additional hour of travel time decreases WTP for a ski pass by €37.3.

[a] Willingness to pay is the ratio of the coefficients in Table 5 ($b_x/-b_{\text{price}}$). Values computed from non-significant coefficients ($p > 0.1$) are printed in *italics*.

individual specific terms ε_{jt} are independently distributed, each with an extreme value distribution. The attributes of the opt-out option were coded with zero values (Kontoleon and Yabe, 2004, p. 18). Based on the choice model parameters we computed the implicit willingness to pay (WTP) for the attributes. Finally, we applied the same multinomial logit specification to explore the implicit willingness to pay for ski area attributes for different customer groups (income, age, living distance to the next ski destination, frequency of skiing per year) in different information treatments.

5. Results

5.1. Descriptive statistics

The respondents were from the following countries (in percent of the sample): Germany, Austria and German speaking part of Switzerland (85), France and French speaking part of Switzerland (8), England (3), Italy (1), Scandinavia (1) and others (2). There were 34% females and 66% male participants. The average age was 37 years, and 75% were Alpine skiers and 15% snowboarders. Based on a comparison with other studies, the sample is representative of the actual population of skiers at least in terms of age, gender, and the kind of snow sports. A sample of 540 Austrian skiers (Unbehaun et al., 2007) showed a gender distribution of 53% male and 47% female, with 74% in an age between 25 and 50 years and 75% Alpine skiers and 8% snowboarders. In a sample with 1825 participants in Austria (Jais, 2001) the average age was 37 years with 38% females and 62% males. Bürki (2000) reports 59% males and 41% females for a Swiss region. A survey for Germany shows 59% male and 41% female skiers and boarders. The mean age in that study is in the age class 30–39 years (Allensbach, 2007). Further statistics describing the respondent sample are presented in Table 3. Chi-square and Kruskal–Wallis test statistics showed that the information treatment groups were not significantly different (at $p < 0.05$) in any of the examined characteristics.

5.2. Effect of NGO and consumer information on destination choices

Table 4 presents the distribution of choices among the options “ski area A” (A), “ski area B” (B), and “none” (omitted in the table) for each choice set and information treatment. Each line represents

an independent test of the null hypothesis that the third-party information did not matter for choice behavior. Given 42 choice sets with positive or negative third-party information, about two significant test results (with $p < 0.05$) would be expected even if there were no information effect at all. Relative to the Control, the NGO information significantly affected the choices in 7 out of 19 choice sets with positive recommendations and in 8 of 23 choice sets with negative recommendations. The consumer information affected the choices in 4 out of 19 choice sets with positive recommendations and in 12 of the 23 choice sets with negative recommendations. Although the recommendations were the same under the NGO and the consumer treatment, the NGO and the consumer treatment were *both* significantly different from the control in only 7 of these choice sets.

The direction of the effects was in all cases as expected; a positive recommendation always increased the probability that the respective option was chosen, and a negative recommendation of both options always increased the probability that the opt-out option was chosen. The NGO recommendations changed the probability that a particular option was chosen by up to 27.2% points relative to the control (Table 4; ski area B in choice set 1, increased from 54.6% to 81.8%). Similarly, the consumer recommendations changed the probability that a particular ski area was chosen by up to about 28% points relative to the control treatment (Table 4; ski area B in choice set 49, decreased from 69.7% to 42.0%; ski area A in choice set 55, decreased from 42.9% to 15.7%).

In six of the 42 choice sets, the effect of the NGO information was significantly different from the effect of the consumer information (last column in Table 4), indicating that the respondents' perception (and use) of the recommendations depended on the sources of these statements. In three of these cases (choice sets 1, 2 and 5) the effects of positive recommendations were stronger (more strongly increased the probability that the recommended options were chosen) when the recommendations came from the NGO rather than from consumers. In two cases (choice sets 21 and 24) the effects of negative recommendations of both options were stronger (more strongly increased the probability that neither option was chosen) when the recommendation came from consumers rather than from an NGO. Hence, it appears that the positive recommendations were more effective when they came from an NGO, while negative recommendations were more effective when they came from consumers.

In principle, the information treatments could also affect choices in cases with "no recommendation". The information treatment, which is applied at the level of the respondents, could change the respondents' mindset and thus the perception and evaluation of the options in *any* choice set, even if none of the options is negatively or positively recommended. We find some evidence of such an effect. Out of 22 choice sets with "no particular recommendation", three were significant.

5.3. Willingness to pay for sustainability attributes

Table 5 presents choice model coefficients for each information treatment. Most of the attributes (or dummies for specific levels) were highly significant under each treatment. The exceptions were ALT1 (snow-independent downhill skiing), EXP1 (no expansion of ski area) and GROOM (100% groomed); see Table 6 for reference levels. From the estimated choice model coefficients we calculated the implicit willingness to pay for the destination attributes (Table 6). While these WTP values should be interpreted with caution (see Section 6), they are useful to further explore the pattern of results.

Most of the sustainability attributes (HOURS, RENEW, ALT1/2/3, FLEX, EXP2/3) indicate a preference for the more sustainable alternatives under each treatment. Moreover, for each of these attributes except ALT1, the willingness to pay increases from the control to the consumer information and to the NGO information treatment. In contrast, the results for the "primary" sustainability attributes that may most decisively affect the skiing itself, including "size of the ski area" (SMALL), "percentage groomed" (GROOM), and "snow guarantee" (SNOW) do not indicate a preference for the more sustainable alternatives. Interestingly, with these attributes, willingness to pay for the *less* sustainable alternatives even increases from the control and consumer information to the NGO information treatment (Table 6). This pattern suggests that the respondents followed the recommendations in evaluating those "softer" or "secondary" sustainability attributes, which do not affect the quality of the skiing but insisted all the more on snow guarantee and fully groomed slopes.

Table 7
Willingness to pay for ski area characteristics in different respondent subsamples.[a]

	Income (€ per month)				Age (years)			Travel distance (h)			Skiing per year (weeks)	
Attribute	<1000	1000–2000	2000–4000	>4000	<31	31–50	>50	<2	2–8	>8	up to 2	>2
N (choices)	2992	4424	7976	2760	6760	8952	3232	11,936	6201	1122	10,545	8776
HOURS	−21.4	−15.8	−25.3	−24.0	−16.1	−32.2	−15.9	−25.3	−16.3	−10.9	−30.0	−14.3
SMALL	−61.3	−44.4	−75.9	−63.8	−51.2	−90.9	−36.4	−63.1	−59.8	−46.9	−80.3	−45.3
RENEW	99.0	84.5	97.5	85.0	76.6	117.0	92.8	108.8	64.5	99.4	124.8	65.2
Alternatives												
ALT1	*−2.4*	*−4.6*	*−1.6*	−29.2	*0.1*	*−11.4*	−18.1	*−3.9*	−9.5	*−20.9*	*−9.3*	*−6.0*
ALT2	−72.1	−61.7	−75.5	−111.9	−54.5	−102.1	−80.7	−80.9	−58.3	−103.2	−98.2	−55.3
ALT3	*−26.8*	−20.7	−63.5	−87.0	−19.3	−88.4	−51.0	−63.9	−23.2	*−25.5*	−62.7	−33.1
FLEX	31.7	44.9	61.6	63.7	40.4	74.5	42.6	53.5	49.4	43.6	64.6	40.9
Expansion												
EXP1	23.6	*8.9*	*3.7*	*12.8*	8.9	16.9	*0.1*	9.9	10.0	*−2.0*	13.4	*5.6*
EXP2	−27.8	−29.6	−32.1	−6.2	−17.7	−36.1	1.2	−43.5	*−6.5*	−47.9	−42.3	−18.9
EXP3	−57.0	−42.7	−73.4	−57.5	−44.0	−83.5	1.6	−71.1	−36.9	−86.4	−84.6	−39.4
GROOM	*5.9*	*−1.2*	11.3	*14.6*	10.2	*1.2*	−0.4	*−2.2*	16.5	47.1	21.7	*−2.1*
SNOW	42.8	29.3	*45.4*	46.7	34.5	56.1	−0.5	42.0	35.0	31.1	47.8	31.8

[a] See Table 6.

Table 8
Willingness to pay for ski area characteristics for different income x treatment groups.[a]

	Income × information treatment group							
	<€1000		€1000–2000		€2000–4000		>€4000	
	Cons.	Control	Cons.	Control	Cons.	Control	Cons.	Control
N (choices)	984	992	1496	1536	2752	2496	976	928
HOURS	−13.7	−15.1	−25.4	−9.6	−22.1	−17.8	−15.0	−21.9
SMALL	−39.0	−44.7	−78.6	−31.7	−67.3	−39.7	−36.6	−65.2
RENEW	93.3	57.8	107.0	65.8	99.8	58.6	41.5	53.0
Alternatives								
ALT1	*9.9*	*−13.6*	−34.0	*7.4*	*−11.7*	*−6.4*	−26.1	*−7.0*
ALT2	−79.2	−34.0	−121.4	−22.8	−98.2	*−16.7*	−75.1	−90.7
ALT3	*−10.9*	*−32.6*	−94.4	*3.0*	−69.6	−31.4	−50.2	−108.6
FLEX	19.1	*12.5*	73.2	39.3	52.8	52.4	54.1	49.9
Expansion								
EXP1	*0.7*	*13.2*	*21.6*	*−10.2*	*5.9*	*7.2*	31.6	*−6.6*
EXP2	*−17.9*	*−11.5*	−91.7	*−11.5*	*−17.5*	−28.6	*20.1*	*−11.5*
EXP3	−52.9	−35.2	−74.6	*−13.9*	−65.8	−32.3	*−11.7*	−56.0
GROOM	*2.0*	*11.3*	*−6.7*	*4.6*	*13.4*	*1.2*	*2.2*	*15.3*
SNOW	26.7	42.3	40.1	22.8	38.4	37.1	27.4	53.1

[a] See Table 6.

5.4. *WTP for sustainability attributes disaggregated by socioeconomic groups*

To explore the effects of sustainability information on different customer groups we also computed the WTP values shown in Table 6 for subsamples with different socioeconomic characteristics (Table 7). These results show that the preferences are heterogeneous among the examined groups. For instance, the respondents above the age of fifty did not have a positive WTP for snow guarantee and fully groomed slopes.

Furthermore, we examined how the information treatments (consumer information *vs.* control) interacted with different socioeconomic characteristics such as income (Table 8). The latter results suggest that communication of this type may have different effects on different customer groups. For instance, when the consumer information was available, travel distance became more important for the medium income groups, smaller ski areas became more desirable for the lowest and highest income groups, use of electricity from renewable sources became more important for the low to medium income groups, and snow guarantee became less important for the lowest and highest income groups. Due to space limitations, we cannot describe and discuss these results in detail. Additional analysis and discussion is provided in Luthe (2009). The general result is that third-party information mattered and its effect depended on individual characteristics of the respondents.

6. Discussion

Our key findings are three-fold. First, we identify a demand for specific – but not all – sustainability attributes of winter tourism services. Second, the demand for these attributes was significantly affected by third-party information in the form of researcher-defined NGO and consumer recommendations. Third, the acceptance of sustainability attributes may differ among different customer groups. Several aspects of these findings warrant further discussion.

The results suggest a positive willingness to pay for many “secondary” sustainability attributes of winter tourism services. Interestingly, however, those “primary” sustainability attributes which may negatively affect the skiing experience itself – notably a decreased snow guarantee and only partial grooming – were evaluated negatively even when NGO or consumer recommendations were provided (see Section 5.3 and Table 6). These findings indicate the limits of third-party information in promoting sustainable winter tourism. Nevertheless, there are specific groups of customers, including those above the age of 50 years, who did not evaluate those attributes negatively. From the perspective

of destination operators, the disaggregated results thus suggest opportunities for offering specific more sustainable winter tourism services to specific target groups.

While the overall pattern of results agrees well with intuitions, the absolute willingness-to-pay values may appear high (see Tables 6–8). This finding could indicate that some individuals perceived the issue rather as citizens than as consumers and placed little weight on the price attribute. Additional reasons for the high willingness-to-pay values may include possible private benefits from a public-spirited response, a potential self-selection of respondents who are more environmentally or risk aware and a survey period during and shortly after the particularly warm winter of 2006/07, the warmest winter in many locations of the Alps on records at that time (Beniston, 2007). Even though the responses may not be representative for an average year at present, their relevance will increase in the future as with the expected climate change winters such as the one of 2006/07 will occur more frequently.

The key statistical tests in this study are the tests of the information effects on the individual choice sets. They confirm that the respondents did not only use and process the available factual information about the alternative options but also relied on heuristics based on NGO and consumer advice. As noted in Section 4.5, the estimation of choice model parameters and WTP estimates relies on a number of critical additional assumptions. Hence, we see the choice models as a way to explore the pattern of effects rather than a way to obtain reliable estimates of WTP or treatment effects on WTP, and we used only simple multinomial models to explore this pattern. In future work survey approaches using third-party information could be combined with advanced modeling approaches such as mixed logit or error correction models to improve the estimation.

A further caveat regarding the WTP estimates concerns the public-good nature of the sustainability attributes. In the present experiment, the demand for sustainability attributes is examined in the context of private consumption decisions. Although this is a common approach (e.g. Carlsson et al., 2007) willingness to pay for sustainable tourism attributes does not reflect economic values for a clearly defined more sustainable outcome. Any price premiums individuals are willing to pay for the sustainability attributes are not exchange values but merely expressing a willingness to *contribute* towards more sustainable outcomes. Hence, while markets involving externalities may play an important role in the transition towards a more sustainable economy, they may not be ideally suited as institutions through which values of externalities can be expressed and measured (Vatn, 2005).

In spite of the well-known influence of third-party information in consumer choices about unfamiliar products, the role of such information in demand analyses or preference surveys has only recently been addressed (Huffman et al., 2007; Macmillan et al., 2002; Schläpfer et al., 2008). The present study extends the findings of previous research to choices in the context of a transition towards more sustainable services where relevant third-party information may be provided by environmental NGOs or other consumers. The significant effects of the consumer and NGO advice confirm the role of heuristic decision-making based on third-party information and indicate that this role warrants further attention in survey research.

In experiments, third-party information can be consciously controlled in order to predict choice behavior in alternative social information contexts, which may be particularly relevant for sustainable product design. An interesting potential of such experiments would be the study of effects of marketing in cooperation with environmental NGOs or consumer organizations. Even in the present experiment, the result could be tentatively interpreted in terms of marketing effects. They suggest that the demand for more sustainable winter tourism destinations could be enhanced through a marketing involving a sustainability rating by environmental NGOs, consumer organizations, or other third parties.

7. Conclusions

Innovations in the tourism industry could play an important role in the transition to more environmentally sustainable winter tourism, but the lack of information on market demand for alternative services has been a barrier for the transition. In this study we examined preferences for sustainability attributes of winter tourism services in a typical situation where potential innovators are uncertain about future demand, where potential consumers do not have well defined preferences over alternatives, and where consumer choices may therefore heavily rely on information from third-parties

such as consumer organizations. Our findings indicate that there exists a demand at least for some "secondary" sustainability attributes of winter tourism services if such services are available and particularly if the sustainability concern is communicated through third-party information. This insight in turn suggests that there are opportunities for ski resorts that react to this demand by developing more sustainable business models and tourism services and by raising the visibility of the sustainability issue through marketing. Conversely, the findings may indicate that third-party information could become a risk and a push-factor for supply side actors not engaging into environmental innovations. Nevertheless, the findings also suggest that the "primary" sustainability attributes which may negatively affect the skiing experience itself – a decreased snow guarantee and only partial grooming – are evaluated negatively even in the presence of sustainability information from third parties.

Complementary research on the supply side of the market for winter tourism suggests that beliefs, trust and partnering among the stakeholders (Abegg et al., 2008; Luthe, 2009) as well as further corporate determinants (Hoffmann et al., 2008) and governance models (Beritelli et al., 2007) are critical for innovation. Together with these factors, information on the evolving preferences for sustainable consumption appears to be a promising and important ingredient in the transition towards a more sustainable winter tourism industry.

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Appendix A.

See Tables A.1 and A.2.

Table A.1
Destination profiles.

No.	TYPE	RENEW	FLEX	ALT	HOURS	EXP	GROOM	SNOW	PRICE
1	Small	0	NO	ALT4	>8	EXP1	100	6	58
2	Large	100	NO	ALT1	2–4	EXP4	50	3	58
3	Small	0	NO	ALT2	>8	EXP1	50	6	34
4	Small	100	YES	ALT4	4–8	EXP1	50	5	58
5	Small	0	YES	ALT4	>8	EXP3	100	3	58
6	Small	100	NO	ALT4	<2	EXP4	100	4	58
7	Large	0	NO	ALT2	>8	EXP4	100	6	58
8	Large	100	NO	ALT2	<2	EXP1	50	6	58
9	Large	100	NO	ALT1	4–8	EXP1	100	3	42
10	Small	100	NO	ALT1	<2	EXP1	50	5	34
11	Small	100	NO	ALT1	>8	EXP1	50	3	58
12	Large	100	YES	ALT1	>8	EXP4	100	3	42
13	Small	0	YES	ALT1	<2	EXP1	50	6	58
14	Large	100	YES	ALT4	<2	EXP3	100	6	58
15	Small	0	YES	ALT4	<2	EXP4	50	4	58
16	Large	0	NO	ALT4	2–4	EXP1	100	3	58
17	Small	0	YES	ALT4	>8	EXP4	50	5	34
18	Large	0	NO	ALT1	4–8	EXP1	50	3	34
19	Large	100	YES	ALT4	<2	EXP2	100	3	34
20	Large	100	YES	ALT2	>8	EXP1	50	3	34

Table A.1 (*Continued*)

No.	TYPE	RENEW	FLEX	ALT	HOURS	EXP	GROOM	SNOW	PRICE
21	Small	0	NO	ALT1	>8	EXP4	100	5	34
22	Small	0	NO	ALT2	<2	EXP1	100	4	42
23	Large	100	YES	ALT4	>8	EXP4	100	6	34
24	Large	0	YES	ALT3	2–4	EXP4	50	3	42
25	Small	100	YES	ALT1	>8	EXP4	50	5	58
26	Large	100	NO	ALT4	2–4	EXP1	100	6	42
27	Small	100	NO	ALT1	2–4	EXP3	100	6	58
28	Small	100	NO	ALT2	4–8	EXP4	50	3	34
29	Large	0	NO	ALT4	2–4	EXP4	50	6	58
30	Small	100	YES	ALT1	2–4	EXP3	50	3	42
31	Large	100	NO	ALT1	>8	EXP3	50	6	42
32	Large	100	NO	ALT4	4–8	EXP3	50	4	34
33	Small	0	YES	ALT1	>8	EXP1	100	4	42
34	Large	100	NO	ALT3	>8	EXP3	100	4	58
35	Large	0	NO	ALT4	>8	EXP1	50	4	50
36	Large	0	YES	ALT1	4–8	EXP4	50	6	34
37	Small	100	YES	ALT2	<2	EXP1	100	3	58
38	Small	0	YES	ALT4	4–8	EXP1	50	3	34
39	Large	100	YES	ALT2	<2	EXP2	50	5	50
40	Small	100	NO	ALT4	>8	EXP4	50	6	50
41	Large	0	YES	ALT1	4–8	EXP4	100	4	58
42	Small	0	NO	ALT1	2–4	EXP4	100	3	58
43	Large	0	YES	ALT3	4–8	EXP1	100	5	50
44	Large	100	YES	ALT1	<2	EXP1	100	6	50
45	Small	0	NO	ALT3	4–8	EXP2	50	4	58
46	Large	100	YES	ALT4	4–8	EXP4	50	3	58
47	Large	0	NO	ALT1	>8	EXP4	50	3	50
48	Large	0	YES	ALT1	4–8	EXP1	50	3	58
49	Small	0	YES	ALT3	4–8	EXP4	100	6	50
50	Large	100	NO	ALT2	<2	EXP4	100	6	34
51	Small	100	NO	ALT4	<2	EXP2	50	3	50
52	Small	0	YES	ALT1	<2	EXP3	100	3	34
53	Large	0	YES	ALT3	>8	EXP2	50	6	58
54	Small	0	NO	ALT1	<2	EXP4	50	6	42
55	Large	100	YES	ALT1	>8	EXP1	100	6	58
56	Small	100	NO	ALT4	>8	EXP1	100	3	34
57	Large	0	NO	ALT4	2–4	EXP4	100	3	42
58	Large	0	NO	ALT1	<2	EXP2	100	6	50
59	Small	100	YES	ALT4	4–8	EXP4	100	4	42
60	Small	0	NO	ALT4	<2	EXP3	100	6	34
61	Large	0	NO	ALT2	>8	EXP2	100	3	34
62	Small	100	YES	ALT2	>8	EXP2	100	6	34
63	Large	0	YES	ALT4	4–8	EXP1	50	6	42
64	Small	100	YES	ALT4	<2	EXP4	50	6	34

Table A.2
Wording of the NGO and consumer recommendations.

Treatment "NGO information"	
Disapproval	"A reputable environmental foundation recommends choosing neither one of the ski areas A or B. Visiting any of the two ski areas can harm the environment significantly taking into account the sum of all features. Basic requirements of sustainable development are not met by these two ski areas."
Recommendation	"A well-known environmental foundation recommends choosing ski area A (B). Visiting ski area A (B) has the least harmful effects on the environment. In a direct comparison ski area A (B) is more sustainable."
Neither	"A well-known environmental foundation does not recommend or disapprove any of the two ski areas because there is not sufficient information on which to base a decision."

Table A.2 (*Continued*)

Treatment "Consumer information"	
Disapproval	"8 out of 10 guests recommend choosing neither one of the ski areas A or B. Visiting any of the two ski areas can harm the environment significantly taking into account the sum of all features. Basic requirements of sustainable development and service are not met by these two ski areas."
Recommendation	"8 out of 10 guests recommend choosing ski area A (B). Visiting ski area A (B) has the least harmful effects on the environment and the best service. In a direct comparison ski area A (B) is more sustainable."
Neither	"There is no information concerning recommendations or disapprovals for any of the two ski areas by past visitors."

References

Abegg, B., 1996. Klimaänderung und Tourismus—Klimafolgenforschung am Beispiel des Wintertourismus in den Schweizer Alpen. Final report of the NFP 31. Zurich, Switzerland.

Abegg, B., Kolb, M., Sprengel, D., Hoffmann, V.H., 2008. Klimawandel aus der Sicht der Schweizer Seilbahnunternehmer. In: Bieger, T., Laesser, C., Beritelli, P. (Eds.), Jahrbuch der Schweizerischen Tourismuswirtschaft. Institute for Services and Tourism, University of St. Gallen, Switzerland, pp. 73–83.

Allensbach, 2007. Allensbacher Markt- und Werbeträgeranalyse. Allensbach, Institut für Demoskopie Allensbach.

Alvarez-Farizo, B., Hanley, N., Barberan, R., Lazaro, A., 2007. Choice modeling at the "market stall": individual versus collective interest in environmental valuation. Ecological Economics 60 (4), 743–751.

Alvarez-Farizo, B., Gil, J.M., Howard, B.J., 2009. Impacts from restoration strategies: assessment through valuation workshops. Ecological Economics 68 (3), 787–797.

Ariely, D., Loewenstein, G., Prelec, D., 2003. Coherent arbitrariness": stable demand curves without stable preferences. Quarterly Journal of Economics 118 (1), 73–105.

Beniston, M., 2007. Entering into the "greenhouse century": recent record temperatures in Switzerland are comparable to the upper temperature quantiles in a greenhouse climate. Geophysical Research Letters 34, L16710, doi:10.1029/2007GL030144.

Beritelli, P., Bieger, T., Laesser, C., 2007. Destination governance—using corporate governance theories as a foundation for effective destination management. Journal of Travel Research 20, 1–12.

Bürki, R., 2000. Klimaänderung und Anpassungsprozesse im Wintertourismus. Report of the Ostschweizerische Geographische Gesellschaft, vol. 6. St. Gallen, Switzerland.

Carlsson, F., Frykblom, P., Lagerkvist, C.J., 2007. Consumer benefits of labels and bans on GM foods—choice experiments with Swedish consumers. American Journal of Agricultural Economics 89, 152–161.

CIPRA, 2006. Wintertourismus im Wandel. Skifahren in Ewigkeit Amen? Alpenkonferenz: Konkrete Maßnahmen für das Klima. CIPRA, Schaan, Liechtenstein.

Druckman, J.N., 2001. Using credible advice to overcome framing effects. Journal of Law, Economics & Organization 17, 62–82.

Elsasser, H., Bürki, R., 2002. Climate change as a threat to tourism in the Alps. Climate Research 20, 253–257.

Endler, C., Oehler, K., Matzarakis, A., 2009. Vertical gradient of climate change and climate tourism conditions in the Black Forest. International Journal of Biometeorology 54 (1).

Falk, M., 2010. A dynamic panel data analysis of snow depth and winter tourism. Tourism Management 31, 912–924.

Fischhoff, B., 1991. Value elicitation—is there anything in there? American Psychologist 46, 835–847.

Fischhoff, B., 2010. Judgement and decision making. Interdisciplinary Reviews: Cognitive Science 1, 724–735.

Greene, W., 2003. Limdep Version 8.0. User's Manual. Econometric Software, Inc., Plainview, NY.

Gregory, R., Lichtenstein, S., Slovic, P., 1993. Valuing environmental resources—a constructive approach. Journal of Risk and Uncertainty 7, 177–197.

Hamilton, J.M., Maddison, D.J., Richard, S.J.T., 2005. Climate change and international tourism: a simulation study. Global Environmental Change 15, 253–266.

Hardin, R.H., Sloane, N.J.A., 2003. Operating manual for gosset: a general-purpose program for constructing experimental designs. Statistics Research Report, AT&T Bell Laboratories, Murray Hill, NJ.

Haward, M.F., Murphy, R.O., Lorenz, J.M., 2008. Message framing and perinatal decisions. Pediatrics 122, 109–118.

Hoffmann, V., Sprengel, D.C., Ziegler, A., Kolb, M., Abegg, B., 2008. Determinants of corporate adaptation to climate change in winter tourism: an econometric analysis. Global Environmental Change 19, 256–264.

Hoyos, D., 2010. The state of the art of environmental valuation with discrete choice experiments. Ecological Economics 69, 1595–1603.

Huffman, W.E., Rousu, M., Shogren, J.F., Tegene, A., 2007. The effects of prior beliefs and learning on consumers' acceptance of genetically modified foods. Journal of Economic Behavior & Organization 63, 193–206.

IPCC, 2007. Climate change: impacts, adaptation and vulnerability. In: Fourth Assessment Report of the IPCC. Cambridge University Press, Cambridge, U.K.

Jacobsson, S., Bergek, A., 2011. Innovation system analysis and sustainability transitions: contributions and suggestions for research. Environmental Innovation and Societal Transitions 1, 41–57.

Jais, R., 2001. Verletzungen im alpinen Skisport unter Berücksichtigung der Entwicklung in der Skitechnologie. Second sample 2001. Faculty of Medicine. Report by the Technical University Munich.

König, U., 1998. Tourism in a warmer world: implications of climate change due to enhanced greenhouse effect for the ski industry in the Australian Alps. In: Wirtschaftsgeographie und Raumplanung, vol. 28, University of Zurich, Switzerland.

Kontoleon, A., Yabe, M., 2004. Assessing the impacts of alternative 'opt-out' formats in choice experiment studies: consumer preferences for genetically modified content and production information in food. Journal of Agricultural Policy Research 5, 1–32.
Konu, H., Laukkanen, T., Komppula, R., 2011. Using ski destination choice criteria to segment Finnish ski resort customers. Tourism Management 32, 1096–1105.
Lauterwasser, E., 1989. Skisport und Umwelt—Ein Leitfaden zu den Auswirkungen des Skisports auf Natur und Landschaft. German Ski Association and Foundation for Safety in Ski Sport, Planegg, DSV-Umweltreihe 1.
Loibl, W., Walz, A., 2010. Generic regional development strategies from local stakeholders' scenarios—the Montafon experience. Ecology and Society 15 (3).
Louviere, J.J., Hensher, D.A., Swait, J.D., 2000. Stated Choice Methods: Analysis and Applications. Cambridge University Press, New York.
Luthe, T., 2009. Vulnerability to global change and sustainable adaptation of ski tourism—SkiSustain. Report series of the Institute of Outdoor Sports and Environmental Science 25. German Sport University Cologne, online URL http://www.tobiasluthe.de/download/SkiSustain_download.pdf.
Luthe, T., 2007. Schneesport und Bildung für eine nachhaltige Entwicklung. Filme, Fachinformationen und Praxistipps zur vernetzenden Kommunikation eines zukunftsfähigen Schneesports in Schule und Verein. German Ski Association and Foundation for Safety in Ski Sport, Planegg, DSV-Umweltreihe 9.
Macmillan, D.C., Philip, L., Hanley, N., Alvarez-Farizo, B., 2002. Valuing the non-market benefits of wild goose conservation: a comparison of interview and group-based approaches. Ecological Economics 43, 49–59.
McFadden, D., 1999. Rationality for economists. Journal of Risk and Uncertainty 19, 73–105.
McKercher, B., Prideaux, B., Cheung, C., Law, R., 2010. Achieving voluntary reductions in the carbon footprint of tourism and climate change. Journal of Sustainable Tourism 18 (3), 297–317.
Miller, G., Rathouse, K., Scarles, C., Holmes, K., Tribe, J., 2010. Public understanding of sustainable tourism. Annals of Tourism Research 37 (3), 627–645.
Mitchell, R.C., Carson, R.T., 1989. Using Surveys to Value Public Goods: The Contingent Valuation Method. Resources for the Future, Washington, D.C.
Mountain Riders, 2007. Diagnostic environnemental de l'activité touristique de la commune de Saint Martin de Belleville en terme d'emissions de gaz à effet de serre: répartition de propositions d'actions. Report. Ademe, Lyon.
Müller, H., Landes, A., Scheurer, R., 2001. Tourismus und Umweltverhalten. Befragung zum Reiseverhalten. Hrsg:Verlag FIF Bern.
Müller, H., Weber, F., 2008. 2030: Der Schweizer Tourismus im Klimawandel. Report. Research Institute for Leisure and Tourism (FIF) University of Berne together with the Swiss National Tourist Office, Bern, Switzerland.
Munro, A., 2009. Bounded Rationality and Public Policy. A Perspective from Behavioral Economics. Springer, Berlin.
OECD, 2007. Climate Change in the European Alps: Adapting Winter Tourism and Natural Hazards Management. Organisation for Economic Co-operation and Development OECD, Paris.
ÖIR, 2007. Stratege. Strategien zur nachhaltigen Raumentwicklung von Tourismusregionen unter dem Einfluss der globalen Erwärmung am Beispiel der Wintersportregion um Schladming Project report. Österreichisches Institut für Raumplanung (ÖIR), Wien. ANr. A 700028.
Payne, J.W., Bettman, J.R., Schkade, D.A., 1999. Measuring constructed preferences: towards a building code. Journal of Risk and Uncertainty 19, 243–270.
Pickering, C., Castley, J.G., Burtt, M., 2009. Skiing less often in a warmer world: attitudes of tourists to climate change in an Australian ski resort. Geographical Research 48 (2), 137–147.
Pröbstl, U., Roth, R., Schlegel, H., Staub, R., 2003. Auditing in Skigebieten. Ein Tool zur ökologischen Aufwertung von Skigebieten, Foundation Pro Natura—Pro Ski, Vaduz, Liechtenstein.
Roth, R., Pazucha, K., Krämer, A., Luthe, T., 2008. Strategic adaptation of the German Mountain Ropeways to Global Change. Working Paper, Institute of Outdoor Sports and Environmental Science, German Sport University Cologne.
Schkade, D.A., Payne, J.W., 1994. How people respond to contingent valuation questions—a verbal protocol analysis of willingness-to-pay for an environmental-regulation. Journal of Environmental Economics and Management 26 (1), 88–109.
Schläpfer, F., Schmitt, M., Roschewitz, A., 2008. Competitive politics, simplified heuristics and preferences for public goods. Ecological Economics 65 (3), 574–589.
Schläpfer, F., 2011. Access to party positions and preference formation: a field experiment. Swiss Political Science Review 17, 75–91.
Scott, D., McBoyle, G., Minogue, A., 2006. Climate change and the sustainability of ski-based tourism in Eastern North America: a reassessment. Journal of Sustainable Tourism, 14.
Scott, D., 2011. Why sustainable tourism must address climate change. Journal of Sustainable Tourism 19 (1), 17–34.
Serquet, G., Rebetez, M., 2011. Relationship between tourism demand in the Swiss Alps and hot summer air temperatures associated with climate change. Climatic Change, doi:10.1007/s10584-010-0012-6.
Simon, H.A., 1957. Models of Man: Social and Rational. Wiley, New York.
Siomkos, G., Vasiliadis, C., Lathiras, P., 2006. Measuring customer preferences in the winter sports market: the case of Greece. Journal of Targeting, Measurement and Analysis for Marketing 14, 129–140.
Slembeck, T., Tyran, J.R., 2004. Do institutions promote rationality? An experimental study of the three-door anomaly. Journal of Economic Behavior & Organization 54 (3), 337–350.
SIf, 2006. Bilanzierung und Reduktion der CO_2-Emissionen in der Landschaft Davos. Eine Machbarkeitsstudie zum Klimaschutz. Schlussbericht zum KTI-Projekt Nr. 7984.1. Swiss Institute for Snow and Avalanche Science SLF, Davos, Switzerland.
Street, D.J., Burgess, L., Louviere, J.J., 2005. Quick and easy choice sets: constructing optimal and nearly optimal stated choice experiments. International Journal of Research in Marketing 22, 459–470.
Sunstein, C.R., Thaler, R., 2003. Libertarian Paternalism is not an oxymoron. The University of Chicago Law Review 70, 1159–1202.
Teich, M., Lardelli, C., Bebi, P., Gallati, D., Kytzia, S., Pohl, M., Pütz, M., Rixen, C., 2007. Klimawandel und Wintertourismus: Ökonomische und ökologische Auswirkungen technischer Beschneiung. WSL Birmensdorf, Switzerland.
Thaler, R.H., Sunstein, C.R., 2008. Nudge. Improving Decisions About Health, Wealth, and Happiness. Yale University Press, New Haven, CT.

Töglhofer, C., Eigner, F., Prettenthaler, F., 2011. Impacts of snow conditions on tourism demand in Austrian ski areas. Climate Research 46, 1–4.
Tversky, A., Kahneman, D., 1974. Judgement under uncertainty: heuristics and biases. Science 185, 1124–1131.
Tversky, A., Kahneman, D., 1981. The framing of decisions and the psychology of choice. Science 211, 453–458.
Uhlmann, B., Goyette, S., Beniston, M., 2009. Sensitivity analysis of snow patterns in Swiss ski resorts to shifts in temperature, precipitation and humidity under conditions of climate change. International Journal of Climatology 29, 1048–1055.
Unbehaun, W., Pröbstl, U., Haider, W.,2007. Trends in wintersport tourism—challenges for the future. In: Conference Proceedings of Leisure Future. Leisure Future, St. Gallen.
Van den Bergh, J.C.J.M., Truffer, B., Kallis, G., 2011. Environmental innovation and societal transitions: introduction and overview. Environmental Innovation and Societal Transitions 1, 1–23.
Vatn, A., 2005. Rationality, institutions and environmental policy. Ecological Economics 55, 203–217.
Witt, U., 2010. Symbolic consumption and the social construction of product characteristics. Structural Change and Economic Dynamics 21, 17–25.
Witt, U., 2011. The dynamics of consumer behavior and the transition to sustainable consumption patterns. Environmental Innovation and Societal Transitions 1, 109–114.
Wolfsegger, C., Gossling, S., Scott, D., 2008. Climate change risk appraisal in the Austrian ski industry. Tourism Review International 12, 13–23.
Yang, J., Wang, C., 2011. Progress in research on the impacts of global climate change on winter ski tourism. Advances in Climate Change Research 1 (2), 55–62.
Zemp, M., Haeberli, W., Hoelzle, M., Paul, F., 2006. Alpine glaciers to disappear within decades? Geophysical Research Letters 33, doi:10.1029/2006GL026319.